$$\text{Induced emf } \varepsilon = -L\frac{di}{dt}$$
$$= -\frac{1.77\times 3}{0.4}$$
$$\varepsilon = -13.275 \text{ V}$$

EXAMPLE 4.11

The self-inductance of an air-core solenoid is 4.8 mH. If its core is replaced by iron core, then its self-inductance becomes 1.8 H. Find out the relative permeability of iron.

Solution

$$L_{air} = 4.8\times 10^{-3}\text{H}$$
$$L_{iron} = 1.8\text{H}$$
$$L_{air} = \mu_o n^2 Al = 4.8\times 10^{-3}\text{H}$$
$$L_{iron} = \mu n^2 Al = \mu_o \mu_r n^2 Al = 1.8\text{H}$$
$$\therefore \mu_r = \frac{L_{iron}}{L_{air}} = \frac{1.8}{4.8\times 10^{-3}} = 375$$

4.3.3 Mutual induction

When an electric current passing through a coil changes with time, an emf is induced in the neighbouring coil. This phenomenon is known as **mutual induction** and the emf induced is called mutually induced emf.

Consider two coils which are placed close to each other. If an electric current i_1 is sent through coil 1, the magnetic field produced by it is also linked with coil 2 as shown in Figure 4.21(a).

If Φ_{21} is the magnetic flux linked with each turn of the coil 2 of N_2 turns due to the current in coil 1, then the total flux linked with coil 2 $(N_2\Phi_{21})$ is proportional to the current i_1 in the coil 1.

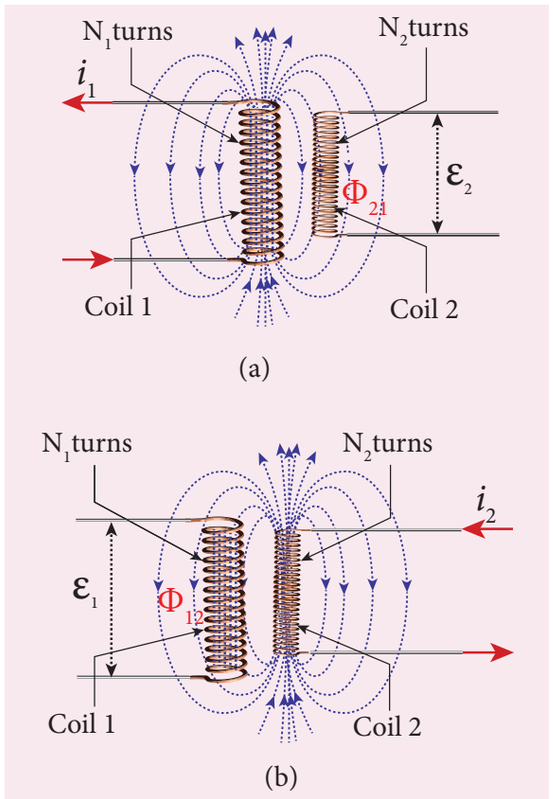


Figure 4.21 Mutual induction

$$N_2\Phi_{21} \propto i_1$$
$$N_2\Phi_{21} = M_{21}\, i_1 \qquad (4.12)$$
$$\text{(or) } M_{21} = \frac{N_2\Phi_{21}}{i_1}$$

The constant of proportionality M_{21} is the mutual inductance or coefficient of mutual induction of the coil 2 with respect to coil 1.

If $i_1 = 1A$, then $M_{21} = N_2\Phi_{21}$. Therefore, **the mutual inductance M_{21} is defined as the flux linkage with the coil 2 when 1A current flows through coil 1.**

When the current i_1 changes with time, an emf ε_2 is induced in coil 2. From Faraday's law of electromagnetic induction, this mutually induced emf ε_2 is given by

$$\varepsilon_2 = -\frac{d(N_2\Phi_{21})}{dt} = -\frac{d(M_{21}i_1)}{dt}$$

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